Notes: when does a Bayesian posterior learn a weak feature?

Finite-n transitions in reduced-rank regression

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These notes collect the reduced-rank part of the project. Statements carry a label: [proved] (with the file holding the proof), [numerical] (with the check), or [heuristic]. Extensions beyond reduced-rank regression are not included here.

1 Setting and the question

Data are n i.i.d. pairs with $y = f(x) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0,1)$, and a model f_w with prior $\varphi(w)$. The excess loss is $K(w) = \frac{1}{2}\|f_w - f\|^2$ with $\|h\|^2 = \mathbb{E}_q h(x)^2$. I work with the population evidence and free energy,

$$Z_n(f) = \int e^{-nK(w)}\varphi(w)\,dw, \qquad F_n(f) = -\log Z_n(f), \tag{1}$$

and with sampled data where stated. For fixed f, $F_n = \lambda \log n - (m-1)\log\log n + O(1)$, with learning coefficient λ and multiplicity m [1].

A weak feature is a decomposition $f = f_0 + \delta g$, $\|g\| = 1$, where f_0 is computed on a larger, more singular set, so that $\lambda_0 = \lambda(f_0) < \lambda_1 = \lambda(f)$. Ignoring the feature costs $\Delta L = \delta^2/2$ per sample, and the signal-to-noise ratio is $s = n\Delta L = n\delta^2/2$.

A phase α with local free energy $F_\alpha = nL_\alpha + \lambda_\alpha \log n + c_\alpha$ overtakes another when $n\Delta L + \Delta\lambda \log n + \Delta c = 0$ [2]. Dropping Δc gives the criterion

$$n\,\Delta L = \Delta\lambda\,\log n, \tag{2}$$

which originates in Sec. 7.6 of Ref. [1] and is applied to stagewise learning in Refs. [4, 3]. It predicts that the feature is learned only once s exceeds $\Delta\lambda \log n$. The question is whether that is right.

To decide when the feature is learned I use a posterior mass: the probability that the fitted amplitude α along g lies within $\delta/2$ of δ. This window marker vanishes at $s = 0$. It measures the projection along the feature, not the full function error. The effective learning coefficient $\lambda_n = n\langle K\rangle$ is not used as a marker, because it rises with s before the posterior moves.

2 One hidden unit: an exact example

Take $f_w(x) = a\,b^\top x$ with $x \sim \mathcal{N}(0, I_M)$, standard Gaussian priors, and truth $f(x) = c^\top x$ with $|c| = \delta$, so $f_0 = 0$, $\lambda_0 = \frac{1}{2}$, $\lambda_1 = M/2$. Completing the square in b gives [proved, exact]

$$Z_n = \int \varphi(a)\,(1 + na^2)^{-M/2}\,e^{-s/(1+na^2)}\,da. \tag{3}$$

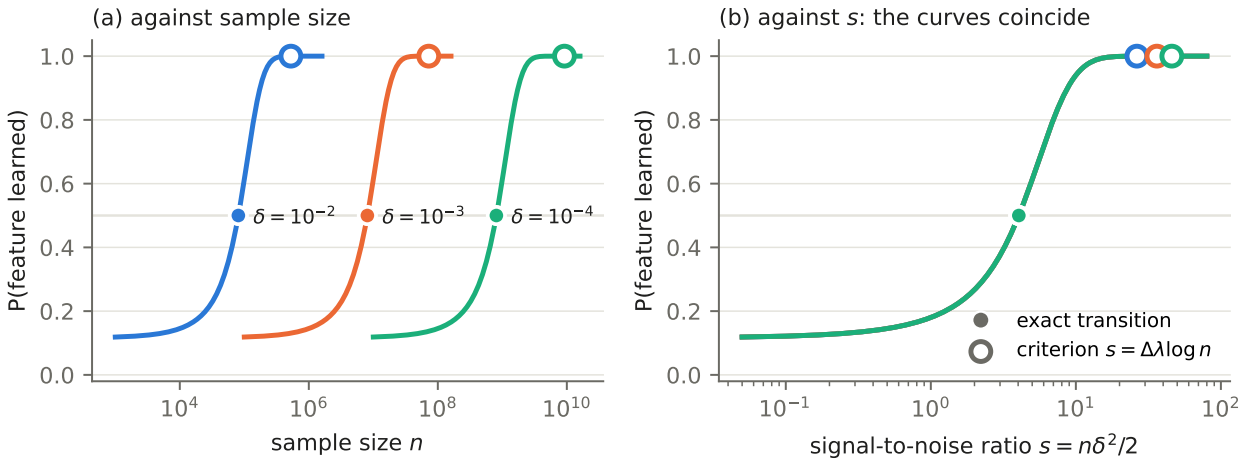


Figure 1: Posterior probability that more than half of the feature is fitted, for $f_w(x) = a\,b^\top x$ with $M = 5$, computed exactly from Eq. (3). Each curve is one truth of strength δ. Filled dots: exact transition. Open circles: Eq. (2). (a) Against n. (b) Against $s = n\delta^2/2$, where the curves coincide.

At fixed s the integrand sits at $a = O(n^{-1/2})$, where $\varphi(a) \to \varphi(0)$. Substituting $t = \sqrt{n}\,a$ removes n from everything but an overall factor:

$$F_n = \tfrac{1}{2}\log n - \log G_M(s) + O(1), \qquad G_M(s) = \int (1+t^2)^{-M/2} e^{-s/(1+t^2)}\,dt. \tag{4}$$

Since $G_M(s) = \mathrm{B}(\frac{M-1}{2}, \frac{1}{2})\,{}_1F_1(\frac{M-1}{2}; \frac{M}{2}; -s) \propto s^{-(M-1)/2}$ at large s, this interpolates as $F_n = \lambda_0 \log n + \Delta\lambda \log(1+s) + O(1)$, uniformly in δ.

The posterior probability that more than half the feature is fitted crosses $\frac{1}{2}$ at $s_* = 4.04$ for $M = 5$, at every n (Fig. 1). Equation (2) puts the transition at $s = 18.4$ for $n = 10^4$ and 36.8 for $n = 10^8$.

Why the criterion fails. [heuristic] For $1 \ll s \ll n$, split Eq. (3) by the size of the output weight. At $|a| \ll n^{-1/2}$ the feature is ignored: free energy $\lambda_0 \log n + s$. At $|a| = O(1)$ it is fitted with all M input weights held to within $n^{-1/2}$: free energy $\lambda_1 \log n + O(1)$. Comparing only these two regions reproduces Eq. (2). The region $n^{-1/2} \ll |a| \ll 1$ also fits the feature, with input weights of order one, next to the branch $\{a = 0\}$ where the input weights are free. Its free energy is $\lambda_0 \log n + \Delta\lambda \log s + O(1)$, lower than the $|a| = O(1)$ region by $\Delta\lambda \log(n/s)$, and the posterior moves into it continuously as s grows. So the fitted phase has $c_\alpha = \Delta\lambda \log \delta^2 + O(1)$: the dropped constant diverges and cancels the $\log n$.

3 Reduced-rank regression

Let $f_w(x) = ABx$ with $A \in \mathbb{R}^{N\times H}$, $B \in \mathbb{R}^{H\times M}$, standard Gaussian priors, isotropic inputs and truth Cx. Write $q = \min(N, H)$, $S = \frac{n}{2}CC^\top$ and $R_n(C) = Z_n(C)/Z_n(0)$. Integrating out B gives [proved, exact]

$$Z_n(C) = \mathbb{E}_A\Big[\det(I_H + nA^\top A)^{-M/2} \exp\Big(-\tfrac{n}{2}\,\mathrm{tr}\,\big[CC^\top (I_N + nAA^\top)^{-1}\big]\Big)\Big]. \tag{5}$$

In singular-value coordinates of A this splits into strata $j = 0, \dots, q$ with exponents $\lambda_j = \frac{1}{2}[jM + (H-j)(N-j)]$; $\lambda_0 = \min_j \lambda_j$ is attained at j_* [5]. The stratum j_* contributes the angular factor

$$\Phi_j(S) = \mathbb{E}_U\,\mathbb{E}_V \exp\big[-\tfrac{n}{2}\|(I - P_U)\,C\,(I - P_V)\|_F^2\big], \tag{6}$$

with U a Haar-random j-frame in $\mathbb{R}^N$ and V a Haar-random $(H-j)$-frame in $\mathbb{R}^M$. For rank-one C, $\Phi_j = {}_2F_2(\frac{N-j}{2}, \frac{M-H+j}{2}; \frac{N}{2}, \frac{M}{2}; -s)$ with $s = n\|C\|_F^2/2$.

Theorem 1 ([proved]; `proof_B9`, Thm. 5)**.** *If j_* is unique, there are constants $C_0, c_0, n_0, \gamma > 0$, depending on N, H, M only, such that*

$$\left|\log R_n(C) - \log \Phi_{j_*}(S)\right| \le C_0\left(\|C\|_F^{2\gamma} + n^{-\gamma/2}\right) \tag{7}$$

for all $n \ge n_0$ and all C of any rank with $\|C\|_F \le c_0$, where $\gamma = \frac{1}{3}\min(1, \lambda_{j_*+1} - \lambda_{j_*}, \lambda_{j_*-1} - \lambda_{j_*})$.

The proof bounds the dominant-stratum integrand above and below by factorised integrals whose ratio tends to one. The bound is uniform in s: it covers $s \to 0$, $s = O(1)$ and $s \to \infty$ with $s \le c_0^2 n/2$. So $-\log \Phi_{j_*}$ is the scaling function of $F_n(C) - F_n(0)$. The error vanishes in the joint small-signal, large-n limit, not at fixed nonzero C.

Its large-s exponent equals Aoyagi and Watanabe's jump $\lambda(1) - \lambda(0)$ for rank-one C [5]. I derive this independently from Eq. (6); it agrees in exact arithmetic for all 8000 triples with $N, H, M \le 20$ [numerical]. For rank-one truths, $F_n(C) = \lambda_0 \log n + \Delta\lambda \log(1+s) + O(\log\log n)$ uniformly in $\|C\|_F \le c_0$.

Ties. [proved, `proofs/proof_B9.pdf`, Thm. 9] Two strata tie ($m_0 = 2$) exactly when $N + H + M$ is odd and N, H, M satisfy strict triangle inequalities. For rank-one C,

$$R_n = \Phi(s)\,\frac{\log \frac{n}{1+s} + O(1)}{\log n + O(1)}, \qquad 0 \le s \le n/2. \tag{8}$$

The multiplicity is paid as $\log\log(n/s)$. At fixed s the extra factor is $1 + O(1/\log n)$; for $s = n^\theta$ it tends to $1 - \theta$.

4 Where the transition sits

Table 1 collects transition locations [numerical]. In every case the transition sits at an n-independent s of order one, while Eq. (2) grows with $\log n$.

Model, n	transition s	Eq. (2)
$a\,b^\top x$, $M = 5$, noise-free, all n	4.04	18.4–36.8
$(3,1,3)$, noise-free, $5{\times}10^4$, $5{\times}10^8$	2.72, 2.61	10.9, 20.1
$(8,1,8)$, noise-free, $2{\times}10^5$, $2{\times}10^9$	8.12, 8.01	42.0, 74.2
$(2,1,2)$ tie, noise-free, 10^4, 10^8, 10^{32}	1.75763, 1.64693, 1.57366	4.6, 9.2, 36.8
$(2,1,3)$, sampled data, 10^2, 10^4, 10^8	2.84, 2.74, 2.73	4.6–18.4

Table 1: Transition location: the s at which the window probability reaches $\frac{1}{2}$ (for $a\,b^\top x$, the fitted fraction above $\frac{1}{2}$). Triples are RRR (N, H, M). Sampled data: mean over 64 data sets. The tie row is reproduced by `review_checks/check_calculations.py`, stable under grid doubling to better than 10^{-11}.

In the tie case the transition drifts slowly, from 2.03 at $n = 10^2$ to 1.557 at $n = 10^{128}$. A fit $s_\infty + a/\log n + b/\log^2 n$ gives $s_\infty \approx 1.551$ [numerical]. This drift is consistent with the $1/\log n$ correction in Eq. (8); the evidence-ratio theorem alone does not prove convergence of the window crossing.

The tie factor also accounts for the constant of the fitted phase. Fitting it against $\log \delta^2$ without the factor gives slope 0.62; with it, 0.507 for $n \ge 10^4$, against the predicted $\Delta\lambda = \frac{1}{2}$ [numerical].

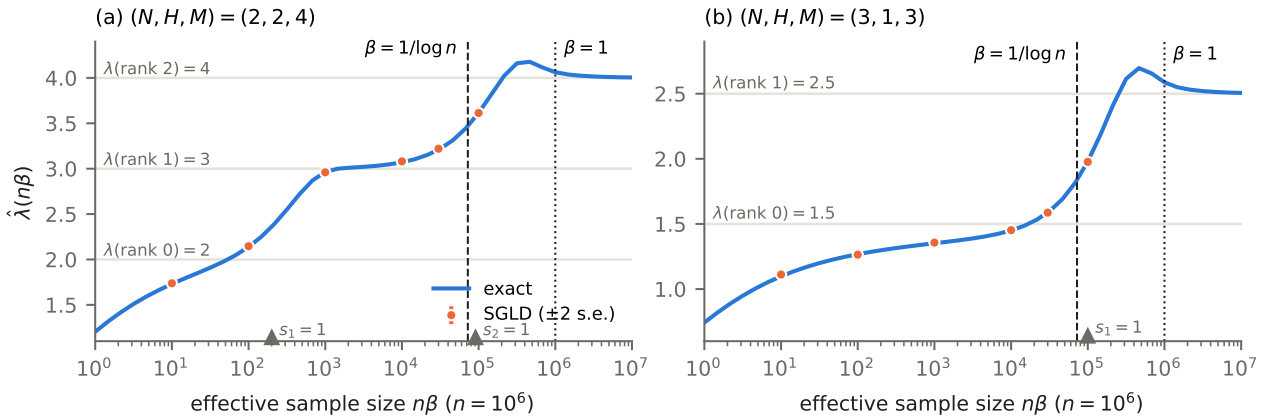


Figure 2: Learning-coefficient estimate of the posterior tempered to effective sample size $n\beta$, RRR at fixed $n = 10^6$, one sampled data set. Line: exact value; dots: SGLD. Gray lines: λ for truths of rank $0, 1, 2$. Triangles: $s_i = n\beta\sigma_i^2/2 = 1$ for the singular values σ_i of the least-squares fit. Dashed: default $\beta = 1/\log n$; dotted: $\beta = 1$. In (b) $\lambda_n(0)$ approaches 1.5 slowly because $(3, 1, 3)$ is a tie case.

Sampled data. For models linear in finitely many features, the w-dependence of the likelihood enters only through the least-squares fit $\hat{C}$ and the empirical second-moment matrix $\hat{\Sigma}$. The empirical evidence is Eq. (1) evaluated at truth $\hat{C}$ in the metric $\hat{\Sigma}$, up to a factor independent of w [proved, exact; checked to relative error 4×10^{-15}]. Data noise therefore shifts the truth by $O(n^{-1/2})$, the same order as the features studied here. The theorem covers the resulting full-rank S, but not the $O(n^{-1/2})$ distortion of the prior from replacing the identity metric by $\hat{\Sigma}$, so the sampled-data transitions are numerical.

5 Temperature and the learning coefficient

At inverse temperature β the unlocalised posterior in this model has effective sample size $n\beta$ [exact, population posterior], so a feature is resolved when $n\beta\delta^2 = O(1)$. Stochastic-gradient Langevin estimators of the local learning coefficient (LLC) sample a tempered posterior, by default at $\beta = 1/\log n$ [6]. A feature with $1 \lesssim s \lesssim \log n$ is then learned by the $\beta = 1$ posterior but not counted at the default temperature [heuristic].

Exact tempered-posterior values for RRR at $n = 10^6$ (Fig. 2) show one step per feature at $n\beta\delta_i^2 = O(1)$, with an overshoot of 10–20% before each plateau [numerical]. At $n\beta = 6.8 \times 10^4$, the grid point closest to $n/\log n$, the estimate is 3.45 for $(2, 2, 4)$ and 1.81 for $(3, 1, 3)$, against 4.06 and 2.59 at $\beta = 1$. SGLD reproduces the exact values to within 0.015. A sweep in β therefore reads off the feature strengths from the step positions, which makes quantitative the observation of Ref. [7] that β sets the resolution of the LLC. A local estimator around a trained parameter adds localisation and baseline subtraction, which this calculation does not include.

6 Related work

Detection of a weak signal at signal-to-noise ratio of order one is known in specific settings, including the one-unit model of Watanabe and Amari, for which the noisy evidence and generalisation coefficient are computed exactly [8, 9], and variational Bayesian matrix factorisation [10]. What these notes add

is the learning-coefficient bookkeeping across all widths of reduced-rank regression, the uniform bound of Eq. (7), the tie correction, and the identification of the dropped constant.

7 Open questions

- Whether the double-scaling form holds beyond reduced-rank regression, and under what geometric condition on the feature. I have partial results and will write them up separately.
- The tie bound at rank two and higher.
- A β-sweep in a small trained network with an independently measured weak feature, to test whether the steps of Fig. 2 survive localisation and sampler error.

Code

The exact evaluator is `code/rrr.py`. `review_checks/check_calculations.py` checks it against an independent integral (agreement 4×10^{-10}), reproduces the tie transitions in Table 1, and runs a regression test for the sampled-data generator.

References

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